

自定义函数

人员

臧晨云 隋湫琛 于昊华 初星熠 霍佳铭 张高 陈之恒 何佩瑜 李景翊 刘子彬 到课

上周作业检查

上周作业链接: <https://cppoj.kids123code.com/contest/4810>

王向东老师周六一点半C++字符串练习4												
刷新												
#	用户名	姓名	编程分	时间	A	B	C	D	E	F	G	H
1	zhanggao	张高	800	77	100	100	100	100	100	100	100	100
2	zangchenyun	臧晨云	800	79	100	100	100	100	100	100	100	100
3	suihaochen	隋湫琛	800	81	100	100	100	100	100	100	100	100
4	huojiaming	霍佳铭	800	90	100	100	100	100	100	100	100	100
5	liuzichen123	刘子彬	700	76	100	100	100	100	100	100	100	
6	chenzhiheng	陈之恒	700	77	100	100	100	100	100	100	100	
7	yuhaozhuo	于昊华	700	78	100	100	100	100	100	100	100	
8	liuyinan	刘奕男	700	78	100	100	100	100	100	100	100	
9	hepeiyu	何佩瑜	700	78	100	100	100	100	100	100	100	
10	lijingxuan	李景翊	700	78	100	100	100	100	100	100	100	
11	chuxingyi	初星熠	700	89	100	100	100	100	100	100	100	0

作业

<https://cppoj.kids123code.com/contest/4950> (作业是 A ~ H 题必做, I 题选做)

课堂表现

今天课上讲了 自定义函数, 同学们课上掌握不错, 课下也要多复习一下, 不然容易遗忘。

课堂内容

判断闰年

```
#include <bits/stdc++.h>

using namespace std;

bool check(int x) {
    if (x%400==0 || (x%4==0&& x%100!=0)) {
        return true;
    }
    else {
        return false;
    }
}
```

```
}

int main() {
    int n;
    cin >> n;
    if (check(n)) {
        cout << "Y";
    }
    else {
        cout << "N";
    }
    return 0;
}
```

质数判断进阶版

```
#include <bits/stdc++.h>

using namespace std;

bool check(long long x) {
    if (x <= 1) return false;
    for (long long i = 2; i*i <= x; i++) {
        if (x%i == 0) {
            return false;
        }
    }
    return true;
}

int main() {
    long long n;
    cin >> n;
    if (check(n)) {
        cout << "Yes";
    }
    else {
        cout << "No";
    }
    return 0;
}
```

编程求 $1 * 2 * 3 * \dots * n$

```
#include <bits/stdc++.h>

using namespace std;
```

```
int f(int x) {
    int res = 1;
    for (int i = 1; i <= x; i++) {
        res *= i;
    }
    return res;
}

int main() {
    int n;
    cin >> n;
    cout << f(n);
    return 0;
}
```

从大到小排序

```
#include <bits/stdc++.h>

using namespace std;

int a[1000005];
bool cmp(int x, int y) {
    return x > y;
}

int main() {
    int n;
    cin >> n;
    for (int i = 1; i <= n; i++) {
        cin >> a[i];
    }
    sort(a+1, a+n+1, cmp);
    for (int i = 1; i <= n; i++) {
        cout << a[i] << " ";
    }
    return 0;
}
```

整数奇偶排序

```
#include <bits/stdc++.h>

using namespace std;

int a[1000005];
bool cmp(int x, int y) {
    if (x%2==1 && y%2==1) {
```

```
        return x > y;
    }
    else if (x%2==1 && y%2==0) {
        return true;
    }
    else if (x%2==0 && y%2==1) {
        return false;
    }
    else {
        return x < y;
    }
}

int main() {
    int n = 10;
    for (int i = 1; i <= n; i++) {
        cin >> a[i];
    }
    sort(a+1, a+n+1, cmp);
    for (int i = 1; i <= n; i++) {
        cout << a[i] << " ";
    }
    return 0;
}
```

不一样的排序

```
#include <bits/stdc++.h>

using namespace std;

int a[1000005];
bool cmp(int x, int y) {
    int a = x, b = y, sum1 = 0, sum2 = 0;
    while (x != 0) {
        sum1 += x%10;
        x /= 10;
    }
    while (y != 0) {
        sum2 += y%10;
        y /= 10;
    }
    if (sum1 != sum2) {
        return sum1 < sum2;
    }
    else {
        return a < b;
    }
}

int main() {
```

```
int n;
cin >> n;
for (int i = 1; i <= n; i++) {
    cin >> a[i];
}
sort(a+1, a+n+1, cmp);
for (int i = 1; i <= n; i++) {
    cout << a[i] << " ";
}
return 0;
}
```

计算2的幂

```
#include <bits/stdc++.h>

using namespace std;

int han(int a) {
    int res = 1;
    for (int i = 1; i <= a; i++) {
        res *= 2;
    }
    return res;
}

int main()
{
    int n;
    cin >> n;
    cout << han(n) << endl;
    return 0;
}
```

求三个数的最大数

```
#include <bits/stdc++.h>

using namespace std;

int han(int a, int b, int c) {
    return max(a, max(b,c));
}

int main()
{
    int a, b, c;
    cin >> a >> b >> c;
```

```
    cout << han(a, b, c) << endl;  
    return 0;  
}
```